Portfolio and Investment Analysis

CAPM Regression and Risk Comparison: Nestlé S.A. and Mitsubishi

This analysis evaluates the systematic risk and market sensitivity of two prominent companies—Nestlé S.A. and Mitsubishi—using the Capital Asset Pricing Model (CAPM) applied to weekly excess returns data. The study employs regression analysis to assess the relationship between company performance and market movements, providing quantitative insights into risk characteristics and portfolio implications.

Objective

Compare systematic risk exposure and market sensitivity using CAPM regression on weekly excess returns data

Methodology

Weekly adjusted closing prices converted to returns with risk-free rate adjustment and excess return calculation

Metrics

Beta coefficient, R-squared, statistical significance, standard deviation, and Sharpe ratio analysis

Regression Framework and Data Processing

Weekly adjusted closing prices were collected for both companies and transformed into weekly returns using the standard formula:

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

The weekly risk-free rate was derived from the annual Treasury yield (approximately 4%) adjusted for the weekly frequency:

$$R_{f,weekly} = \frac{0.04}{52}$$

Excess returns were computed by subtracting the risk-free rate from both company and market returns. The CAPM regression model was estimated using ordinary least squares:

$$R_i - R_f = \alpha + \beta(R_m - R_f) + \varepsilon$$

Where β represents systematic risk (market sensitivity), α captures abnormal return (Jensen's alpha), and R^2 indicates the proportion of variation explained by market movements. Statistical tests confirmed model validity through F-tests and coefficient significance assessments.

Results Summary: Key Statistical Findings

Mitsubishi

- Beta coefficient: **1.386**
- Statistical significance: **p < 0.001**
- R-squared: **0.404**
- F-test: **Model significant**
- Alpha: **Not significant**

High market sensitivity with 40% of return variation explained by market movements

Nestlé

- Beta coefficient: **≈0.004**
- Statistical significance: **p > 0.95**
- R-squared: **≈0.000006**
- F-test: **Not significant**
- Alpha: **Not significant**

Minimal market relationship with negligible systematic risk exposure

Interpretation and Risk Characteristics

Mitsubishi: Market-Sensitive Asset

The beta coefficient exceeding 1.0 indicates higher systematic risk relative to the market benchmark. Approximately 40% of return variation is explained by S&P 500 movements, confirming meaningful market dependence. The strong statistical significance (p < 0.001) validates the regression relationship. This profile suggests Mitsubishi behaves as a cyclical, high-beta asset suitable for investors seeking market exposure.

Nestlé: Defensive Characteristics

The near-zero beta and lack of statistical significance reveal minimal systematic risk exposure during the observation period. The regression shows no meaningful relationship with market movements, indicating defensive qualities. This suggests Nestlé's returns are driven by company-specific factors rather than broad market trends, making it a potential portfolio diversification tool and lower-risk holding.

Portfolio Implications and Performance Insights

The risk and performance comparison reveals stark contrasts in systematic risk profiles. Mitsubishi demonstrates higher beta (β > 1) and substantial market dependence, whilst Nestlé exhibits considerably lower market sensitivity. These findings have direct implications for portfolio construction and risk management.

01

Systematic Risk Exposure

Mitsubishi requires higher risk premium compensation; Nestlé offers reduced market volatility exposure

02

Diversification Benefits

Nestlé's low market correlation provides portfolio diversification value; Mitsubishi amplifies market movements

03

Strategic Positioning

Mitsubishi suits aggressive market timing; Nestlé serves as defensive anchor in portfolio construction

Conclusion and Practical Applications

This analysis demonstrates a clear contrast in systematic risk exposure between the two companies. Mitsubishi behaves as a high-beta, market-sensitive asset with 40% of return variation explained by market movements. Conversely, Nestlé exhibits defensive characteristics with statistically insignificant market linkage during the observed period.

The findings illustrate practical applications of regression analysis, risk measurement, and quantitative investment evaluation within portfolio frameworks. Investors can leverage these insights for strategic asset allocation, risk management, and performance benchmarking. The CAPM framework provides a systematic approach to understanding market relationships and informing investment decisions based on empirical evidence rather than subjective assessment.

Key Takeaway: Quantitative risk analysis through CAPM regression enables evidence-based portfolio decisions by systematically measuring and comparing systematic risk exposure across investment alternatives.

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